



HIGHER DIPLOMA IN COMPUTER SCIENCE (NFQ – LEVEL 8)

Project Handbook 2025-26

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Department of Computing & Mathematics

SETU WATERFORD

Table of Contents

1. Project Proposals	3
2. Project Timelines	5
Sample Plan for Semester 5:	6
Design	6
3. Project Submissions	9
Proposal	9
Ethical Approval Checklist	9
Interim Report	9
Showcase Summary	10
Part 1	10
Part 2	10
Final Code	11
Final Report	11
Video	11
Presentation	12
Artefact Usage	12
4. Submitting	12
NDA Policy (subject to amendments)	12
5. Declaration of Authenticity	13
6. Assessment	14
PART A - Core Criteria (85%)	14
PART B - Critical Self-Review (15%)	15
Assessment Process	15
Assessment Examiners	16
7. Project Supervision	16
Project Meetings	16
8. Project Finishing Date	17
9. FINAL MARK - How does Work Placement, Project, and Certification come together?	18
Final Marking for the module:	18
Appendices	19
APPENDIX 1: Project Proposal Outline	19
APPENDIX 2: Video Guide	20
Video Software Recommendations:	20
Submitting	21
APPENDIX 3: Presentation Guide	24
Planning your Presentations:	24
Typical Timing	24
Tips	24
APPENDIX 4: Sample Final report elements	25
Cover Page	25
Table of Contents – Samples	26
APPENDIX 5: Declaration	35
APPENDIX 6: Academic Writing Workshops	36
Sample Schedule	36
APPENDIX 7: Sample Ethics Checklist for School of Science & Computing	37

1. Project Proposals

Please join the **#project** channel in **Slack** to receive all announcements including deadlines. It is your responsibility to check-in regularly for announcements.

The purpose of the project is:

- To provide a Capstone for learning from the course
- Integrate knowledge from multiple modules
- Investigate new domains, technologies or procedures
- Document and record your work placement
- Showcase skills and achievements
- Formally research a new technical domain

You are invited to make project proposals on an individual basis. The proposal may reflect a deep personal interest, the interest of your supervisor or aspects of the technology and business processes of your work placement. It may be useful to conceive a project as a solution to a problem. Representative project examples include:

i. **Work Based Project / Package**

A project allocated by your work placement supervisor, representing a reasonably independent piece of work you have been implementing on your placement:

- A standalone **application** (e.g. web app, mobile app, native or hybrid)
- A **component** of a larger system
- A new or improved **workflow / business process** serving some need in the workplace (e.g. CI/CD continuous integration / continuous deployment)

ii. **Independent Project**

Unrelated to work placement, consisting of an implemented software system designed to explore or consolidate knowledge in a specific domain

iii. **Research Paper**

Exploring a specific topic or set of topics and developed to conference paper standard. May or may not be related to work placement. This project type is **rare** as the course is focused on software development implementation.

The project may be directly aligned with your work placement opportunity, or it may be independent of it. This may depend on the disposition of the industry providing the placement, your personal goals and interests, and the advice of your project supervisor.

You will have **two meetings** in semester 4 to

- discuss your project proposal,
- complete an ethical checklist,
- be advised on next steps.

Before deep diving into a project you must **complete the SETU School of Science Ethical Checklist with the project coordinator**. A key component for all projects is NOT to use sensitive data (you will need to create a test data set)

- If you are **working in tech**,
 - You have choice of work based or independent project.
 - We recommend **work based project**. Your workplace will usually allow you time to work on it as part of your day-to-day work.
 - As part of the process you will need to submit your job spec to the Placement Coordinator (Joan Mangan) which will be reviewed.
- If you are **not working in tech**,
 - You are by definition doing an **independent project**.
 - You are incentivised to enhance your learning experience by undertaking **certification programmes** that we will make available.
 - Project grading for such projects will take this into account. **Project mark is capped at 70% but also floored at 40%** unless you undertake certification (**In other words, you will drop a grade** from the initial marks awarded after the project is presented).
- **Free Certification** options will be made available to all students irrespective of whether you are working in tech or not including RedHat Academy (7 courses) we recommend you complete Sys Admin I, II ,III. AWS certification courses will also be available this year. **Certification** can be undertaken by any student from September to August. **Please take note of the deadline for certification completion (usually third week in August). We don't control or have the power to extend this deadline.**
- In **rare** circumstances the project may take the form of a research project – however this is generally not advised.

2. Project Timelines

Traditionally final year projects run over 2 semesters. You are completing a 30 credit module called ICT Placement Skills. It includes work placement, certification and a project. Work placement is optional. The project is mandatory and must be completed successfully to graduate.

The HDip in Computer Science is an accelerated programme, and the project is accelerated also. However, to assist with balancing your workload, the project extends into a fifth semester and in some cases beyond.

The body of work constituting "The Project" will commence in Semester 4. Requirements, analysis, market & technical research, initial designs, and a sprint plan will be developed prior to your first supervisor meeting in Semester 5.

"The Project" can be divided into 4 phases:

i) Inception Phase:

Includes formation of the project concept,
Two initial meetings (half hour each) with project coordinator,
Draft and Final proposals + (optionally) interaction with workplace mentor.

The Project idea(s) will be discussed with the project coordinator at the first meeting. Ideally you should have a draft ready for each idea in advance. The action from this first meeting is to complete the Final Proposal within 2-3 weeks. A second meeting will review this document and explore and scope the concept further. The proposal is light (about 1-2 pages 500 words approx.)

The action from the second meeting is to complete and submit the Final Proposal within a week or so and then move on to the next phase. **Online Ethical Approval Checklist must be completed with your coordinator after the proposal is finalised.** You will find all materials related to the Ethics Checklist here:
<https://moodle.wit.ie/course/view.php?id=176894>.

The code for this course CM-HDIPCS.

When completed with the project co-ordinator, you should save a copy as pdf.

For now, put down "HDIP-CS PROJECT CO-ORDINATOR as project supervisor when asked

It is inadvisable to proceed with your project without ethical approval.

ii) RAMP Phase – Research, Analysis, Modelling & Planning (Analysis & Design Phase):

With the proposal finalised, carry out user requirements, feature analysis, market analysis (comparing similar products), repo analysis (repos on Github or Bitbucket that address your idea. Maybe you can contribute to these projects or fork them?) & technical research, and initial designs. Choose your methodology, create user stories and organise them into sprints. Create a sprint plan (in Trello/Jira or similar) for the development phase.

Modelling

This might be informed by creating personas for the users of the system. Then creating scenarios/walkthroughs of the system (telling a story about what happens next). In parallel you can create the actual models / diagrams:

- data model,
- system model
- pipeline model
- wireframes for screens
- (consider using Miro, Figma, others)

With all of this done, the customer prioritises what features are needed for a minimal viable product (MVP). Maybe use User Stories for this. Estimate how long it will take for each user story (Manage all this with Trello or Jira or an alternative)

Choose your **Methodology** - write a paragraph on your version of the chosen methodology (software engineering lifecycle/methodology)

Initial **Research / Analysis / Design** 2-4 weeks. Write up your findings and conclude, make your choices early.

Make a **PLAN** for the next 12 weeks -

e.g. use Trello/Jira/Miro to create cards for each task to be done. Estimate time for each on the card. Calculate realistically how much time you are going to spend in each week working on your project. Organise the cards in to stacks for each 2 week sprint (bearing in mind the estimated effort required and the time you have). Track your progress on the cards (RED for not done, Green Done, Amber in progress). Share with your supervisor in January. You have approximately 12-14 weeks to code complete (end of march).

Sample Plan for Semester 5:

6 sprints of 2 weeks

- one for research and analysis (if not completed before hand)
- half for initial designs, screens, data model, walkthrough
- last one left empty for fixes/fancy features

This leaves 3.5 sprints to implement everything (IMPLEMENTATION)

You may need to **rescope** your project - discuss with your supervisor in January.

Use **git** to manage revisions/commits (**daily**). Your git history for the project will be examined. Consider including it in your Final Report.

Design

- screens designs
- walkthrough
- data model

Use pencil and paper to mock up screens. Photograph them. Put each one on a slide. use invisible buttons to jump to slide when click. This will give you an electronic paper prototype. Now write the walkthrough story/stories referring to the labelled screens.

Update as necessary. Now look at the data that is required on each screen, this informs the data model / database.

This work will mostly be completed before and during the Christmas period, ready to be reviewed by your appointed supervisor. Ideally, you should know who this is before end of semester 4.

iii) Development Phase (usually Jan, Feb and March):

Realisation of the project as proposed or amended in agreement with your supervisor. One- or two-week sprints are recommended culminating in a meeting with your supervisor. Communication should be through Slack. Students should keep minutes and action points from each meeting. User stories can be reviewed, and the next sprint planned with your supervisor at each meeting. Students and Staff are advised to use Slack for communications as is the norm on this programme. During this phase students should be updating GITHUB with their code. They should also be working on the Final Report each week. A first draft called the Interim Report is due in mid-February. Seek feedback on this from your supervisor, ideally at the next meeting after submission. A project web page (landing page) should be created and hosted (e.g. on Github or AWS. Don't use a templating service!) which can be amended and republished as the project proceeds.

Assuming 12-14 weeks for development (depending on whether you are working through holidays or not). This equates to 6/7 two-week sprints.

iv) Dissemination Phase:

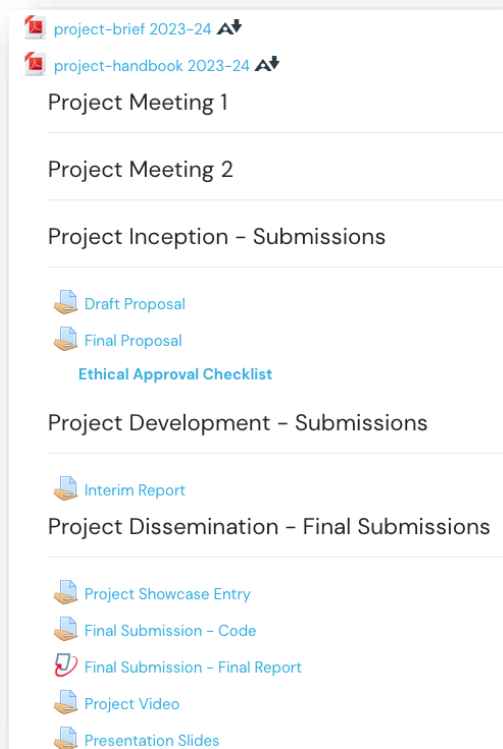
Several deliverables are required for your project.

- **Project Website** (landing page with links to all deliverables)
- A **SINGLE GitHub Project Repo** (which may have links to sub repos)
- Creation of a **Showcase Entry** (project type, name, profile picture, project title(s), abstract, image representing the project, technologies/keywords, links to project website, github, etc.)
- Preparation of **Final Report**.
- **TRELLO/JIRA** board(s) or similar.
- Creation and publishing of a demonstration **video**.
- Project **demonstration/presentation**.
- Delivery of final documentation.
- Presentation of project at workshop 4 (in person or remote)

To help you plan your project, these **milestones** will serve as a useful guide:

Semester	Milestone	Description	When?
4	1	Meeting 1 with Project Coordinator	October
4	2	Draft Proposal	Ideally before Meeting 1. Latest Early November
4	3	Meeting 2 with Project Coordinator	November
4	4	Final Proposal	Ideally before Meeting 2. latest Early December
4	5	Online Ethical Approval Checklist https://moodle.wit.ie/course/view.php?id=176894 <i>see Appendix 7</i>	Early December Ideally at the end of Meeting 2
		RAMP Phase Research, Analysis & Design, Methodology, Plans	Ideally Oct, Nov or before. Latest Dec to Mid Jan
		Individual Project Supervisor Appointed	end of semester / start of next semester
5	6	Interim Report	Mid-February
5	7	Showcase Entry (including poster)	End of March
5	8	Final Project Submission (code)	+1 week
5	9	Final Project Submission (final report) submitted through Turn it In	+1 day
5	10	Video demo of project	+1 week
5	11	Presentation of Final Project to judging panel	+1 week
5	12	Initial Grade of Final Project	+2 weeks
5	13	Presentation of Final Project at http://expo.setu.ie	May/June

3. Project Submissions



During the project's development and realisation, a number of artefacts are created: Please be aware of plagiarism policy, student behaviour policy, use of generative AI policy. As computing students we encourage the use of tools, however they must be acknowledged, cited and used appropriately.

Proposal

This will articulate the project concept and nature and will serve to scope the work of realising the project. It may typically be less than **<2 pages**. For work based projects you need to identify and get permission at this stage for your project from the employer, Flag NDA requirements now. NDA Scope should be restricted. (see NDA Policy below)

Ethical Approval Checklist

In order to ensure your project is viable without requiring formal ethical approval from the university's ethics board, you will need to complete this checklist. This may involve modify the scope or nature of your project. Without successful completion you cannot submit the project. This will usually be completed during Project Meeting 2.

Interim Report

This will be a substantial update on the progress of the project – and can be regarded as an early draft of the final report. It may be **up to 4000 words**, depending on the project type. Completing the RAMP phase will produce the contents expected at this stage. Completing it highlights the work required in completing the Final Report. This report is not marked. The interim report will be present and future tense – where are you now and what will be done in the remaining weeks. Your Supervisor should give you feedback on it and identify areas to be worked on for the Final Report.

Showcase Summary

A showcase is produced each year. Look at last year's book, to better understand its purpose. In order to automate its compilation, we will require several small details which will be requested through an online form or similar. Character limits will be enforced on each field. We request the information in 2 stages:

Part 1 - Immediately after you submit your interim report,

Part 2 – Later, a week before the end of the of your project.

Part 1

- First Name
- Last Name
- Student number (student email address)
- Personal Digital Portrait Photo
- Commercial Title
- Academic Title
- Technology Keywords
- Project landing page (**web url**)
- NDA
- Supervisor's Name

Part 2

- Abstract
- Project image
- Keywords (Tags) / Project Type
 - o (Web App, Workplace Project, DevOps, Open Source, Native Android App, Progressive Web App, Hybrid App, Physical Computing, CICD Pipeline, Testing, AI/ML)
- Link to Code Repo
- Link to YouTube Video

Specifications

- Use a clear up to date personal digital **portrait photograph** with a plain contrasting background. **It should be square width 300 height 400, saved as .jpeg or .png.** Use <https://pixlr.com/> to edit it using the Crop tool width 300, height 400. You should be instantly recognisable from this photo.
- **commercial title (project name)** e.g. "FindThatBook.com". (50 characters combined)
- **academic title** of the project e.g. "distributed library book retrieval system".
- **Project Landing Page. URL** of Project webpage (shorten the link using **bit.ly**). This is a web page which acts as a home/landing page for your project. You can add links to this page as the progress continues. Use github pages, S3, netlify, your own web hosting, wix, whatever you like.
 - o Create a single web page, with name, project, photo, and additional links for the project which will include some or all of the following:

- your proposal, final report, hosted video, presentation slides, link to site or app download, deployment instructions, Slack, Trello, github repos used in the project, link to application, etc.
- **Abstract/Summary** text **500 characters max**, for use in showcase handbook
- A **project image** to represent your project (e.g. 1920*1080). i.e. a visual poster. This could be a screenshot or a composition of many screenshots. System diagram, logos for key technologies used (Quick TIP: capture some screens, crop and paste onto a Powerpoint slide width 1920, height 1080, and export as .png). Treat it like a poster! Use a background or border.
- **Project Type** {web app, mobile app, both, CI/CD, physical computing, AI, OTHER}.
- **Technologies/Keywords** { React, Flutter, Kubernetes, AWS, etc).
- **GitHub Project Repo**. Your project may have multiple repos which you should combine by providing a parent repo. **ONLY SUPPLY ONE REPO!**
- **YouTube Link** to Project Demo (**we will include this later**)
- **Project Poster (which will be displayed at SETU Computing EXPO) – as per project image**
- **Supervisor's name**

Final Code

Your codebase, can be submitted as a **zipped archive** or other repository format (e.g. Github).

Final Report

The final report is **up to 8000 words**. The final report is written in the past tense – what you did. Remember images don't affect your word count. Your final report should have most of the following features: a cover page, declaration of authenticity, acknowledgements, titles, preface, abstract, keywords, glossary, list of abbreviations, TOC, TOF, TOT, appendices, and a bibliography. Use Harvard Referencing style for **citations**. The TOC will vary depending on the project, write to the assessment criteria, and the development of the project. Make sure to include models, analysis, design, iteration plans, implementation details, reflection (what you achieved, learned, future development, problems encountered and solutions).

Use numbered headings and styles, page numbers, **and Harvard referencing**. Use numbered captions on tables and figures. Appendix 4 shows some samples of final report elements.

After the final report deadline, the upload facility will be closed (cut-off) and the reports downloaded and distributed to the panel for review.

Please note you must **cite and reference properly**.

We are looking for your words not an AI generator's words (which must be cited)

Ideally you should submit a draft to your supervisor a week before the deadline to get constructive feedback before final submission.

Video

The **video** must be hosted (e.g. YouTube). Use your student account on YouTube. Make the video unlisted or public. Both the link to the video on YouTube and the original video will be submitted in Moodle. See Appendix 2 for more details on the video.

Presentation

When presenting the final project in person, you will prepare a presentation e.g. Powerpoint. This **presentation** will be submitted just before or straight after your presentation.

Artefact Usage

Artefacts produced and submitted as part of the project (including reports, images, video, code, and slides) may be used for future quality, training, and promotion purposes. Final projects may be archived, as part of the Institute's efforts to provide a repository of the interesting and valuable work carried out by its members and to help future students. Sensitive work can be excluded by way of NDA subject to NDA policy.

If your project is shortlisted for the Best Project Award you will be contacted to decide if you want to give permission for the project artefacts to be reviewed by the award sponsor.

4. Submitting

You will be required to adhere to the milestones outlined in Section 2. All submissions are uploaded to Moodle. Soft copies are required. Hard copies are no longer required.

NDA Policy (subject to ammendments)

In encouraging students to take on work based projects, we understand that some projects may be sensitive. We have used NDAs in the past to facilitate these. However, blanket NDAs for all student work is frowned upon. Where a project is sensitive to a particular company for work / work placement, **if NDAs need to be signed these should be submitted at proposal stage.** The scope of the NDA should be constrained and needs to be discussed early on.

Based on past experience NDAs are requested for five reasons (1. Sensitive data; 2. Sensitive code; 3. Sensitive manager/company/industry; 4. So a student doesn't have to share their work; 5. Because we can!)

Some guidelines:

1. **Student projects can no longer work on live personal / sensitive DATA** – as they fall foul of ethical approval. (They can use anonymised test data for real world systems being developed.) Students will need to use and or produce this test data. NDA not applicable
2. **The project review panel must be able to review and mark CODE produced by the student.** If this is not possible, the work is unsuitable for a university project. Code does not have to be made public. Company code libraries used in the project do not need to be shared/submitted. A private repo is preferred with limited time access to the project panel of three. We will consider alternative arrangements where this is not feasible.
3. We appreciate that software being produced in **some industries may be more sensitive** than others. We will review on a case by case basis.
4. **We won't NDA work just to make it private for a student.**

5. NDAs will be signed if there is good reason, not just because it is a project for a company .

6. Students have to produce artefacts as part of their project.

- Blanket NDAs will prevent any mention of anything to do with the project in Computing Expo etc. We would prefer students to be able to:
 - Describe their project (without necessarily mentioning the company) for their Showcase Abstract
 - Use a more abstract title for their project if this helps
 - Produce and share a poster about the project (possibly more abstract if this helps)
 - Produce a video demo of the project (which can obscure company name etc., for example)
 - Produce a more abstract final report which can refer to private appendices.

When processing the Showcase Handbook entries, we would like to share as many artifacts as possible for each project. We would encourage each company/student to discuss early on, what can be shared and how this is reflected in the individual NDA.

Students still need to submit ALL the deliverables as normal plus additional data as requested, after NDAs are submitted. If covered under NDA they should be flagged. We will respect NDAs and we will keep the details of NDA projects private where NDAs are suitable, and have been submitted and signed in advance.

5. Declaration of Authenticity

On the first inside page of any report, you should have printed out the declaration shown in Appendix 5.

6. Assessment

In preparing your project for assessment, you should consider both core criteria and a critical self-review:

PART A - Core Criteria (85%)

- **Model (15%)**
lightweight, relevant modelling, generally in accordance with a recognised process. For a software project this is most commonly expressed in a subset of UML or a set of appropriate diagrams. For other types of project, a different formalism may be more appropriate. e.g., UML, Architectural Model, Process Flow Charts, Data Models (E-R), UI wireframes etc.
- **Documentation (15%)**
project-related communication including, but not limited to:
 - learning logs (work placement),
 - writing well-constructed formal reports (tables, numbering, Harvard Referencing & citations),
 - sketches of ideas in diagrammatic/written form, wireframes, Trello board (used for plans and progress).
 - Submitting all requirements (draft & final proposal, Handbook entry, Project Web Page, Final Report, Video, Code, Github repo - history/releases).

In keeping with the ethos of the programme, all project communication should be through **#Slack**. Announcements will be made on the Slack **#Project** channel. Communications, interim submissions, documentation, etc. should be exchanged with your advisor in a private channel. Initially, these may be informal photos of notes/diagrams, and may evolve into more formalised minutes as the project progresses. **Trello** would be a good choice for planning. Work Placement **Learning logs** can help. If you are doing a workplace project as part of a team, you should show a breakdown of the elements you worked on, and the percentage that you are responsible for.

- **Implementation (30%)**
based on the modelling and the content of the reports. (i.e., Have you implemented what you said, or something different. If the latter, you should update the documentation before submitting)
- **Level (25%)**
appropriate mix of (a) originality, (b) innovation and (c) complexity. (Think of this as linked to the multipliers used in semester 1 assignments.)

Penalties will be applied for late submissions, exceeding the character count, word count, or maximum durations specified. Penalties will also be applied for not submitting what is requested, in the proper format, or for not citing or acknowledging (plagiarism).

If using generative AI tools, it is advised to include a section evaluating and reflecting on its use.

PART B - Critical Self-Review (15%)

- what you **learned (5%)**
- what you **achieved (5%)**,
and in what direction the project might be taken if more time was available (**future development**)
- **problems** encountered (**5%**)
how they were addressed and solved

In addition, you must attend all meetings with your supervisor. There are no marks allocated for these, but failure to attend, without good cause, will disrupt the assessment procedure.

Assessment Process

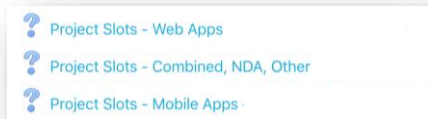
Marking is done at the end of the project.

Project Grade categories are considered (outstanding 85+, excellent 70-84, v.good 60-69, good 50-59, fair 40-49, fail <40)

Before presentations are made :

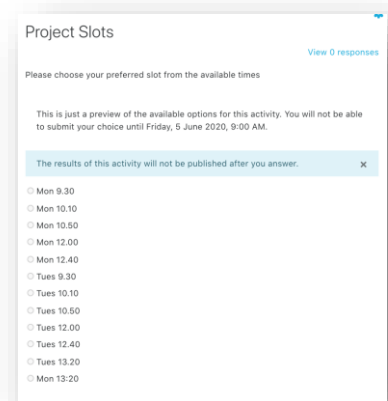
- A handbook will be created grouping the projects by type.
- Presentations will be scheduled, usually no more than six per day.
- A panel of 3+ lecturers will be tasked with reviewing all projects. The panel makeup will be communicated to all in the week before presentations. For large classes, additional panels may be created and each panel will look at a subset of all projects.
- Your supervisor will provide marks to the panel. These are for indicative purposes only.
- All panel members will have viewed your demo video.
- At least 1 member will have read your report in detail.
- At least 1 member may have reviewed your code repos.
- All panel members will be present for your presentation. Your supervisor may attend also.
- After each presentation, the panel member will have an indicative grade or ranking for each project.
- After all presentations have been completed, the panel will meet to discuss the projects in a collegiate fashion, ranking the projects for each category initially starting with highest and lowest working towards the middle. When all project categories have been completed, the panel will break.
- On return, the projects within each category will be merged.
- Then, marks will be applied based on the Project Grade categories.
- The project marks will be shared with all supervisors
- If there is an unacceptable difference between a supervisors suggested mark and the panels allocated mark, the project and mark can be reviewed again.
- Where there are more than 1 panel, there will be an additional step of reviewing and merging the results of each panel.

Assessment takes place at project demos. Typically, a group of similar project types will be scheduled on a given day, at half hour or forty-minute intervals.



A timetable of available slots will be made available for booking on a first come first served basis:

You will be able to book your project slot on Moodle on a first come first served basis (providing you have uploaded all requirements previously).



Assessment Examiners

There are three examiner categories:

- (a) The Supervisor
- (b) Project Panel
- (c) the External Examiners

The Supervisor will suggest a set of marks.

The Project Panel will consider these marks in the context of seeing all projects and will decide the final mark in a collegiate fashion.

The External Examiner will review the marks and may suggest changes.

7. Project Supervision

Each student will be appointed an academic supervisor to provide supervision and advice. The norm will be that you will have a coordinator for 2 meetings in semester 4, and a supervisor appointed for semester 5. It is recommended you discuss the project progress at regular intervals with your supervisor, usually face to face (e.g. Slack/Zoom call). Submit deliverables to your supervisor first a week in advance to allow time for constructive feedback.

Project Meetings

You have a quota of 12 half hour sessions with your supervisor. This may be broken up as needed.

Typically, there may be just 1 or 2 * half hour meetings in the inception phase with the project coordinator.

Most meetings will occur in semester 5 during the development and dissemination phases. Usually, each project student will have a weekly meeting with their supervisor consisting of (12 half hour meetings or 6 one-hour meetings) at the discretion of the supervisor and workplace mentor. He or

she will show his/her progress, emphasis being placed on risk-reduction from week to week as issues associated with (a) project requirements gathering, (b) technologies and (c) skills building are addressed.

For each meeting, **brief minutes and action points** should be taken by the student. Consider creating cards and updating progress using **Trello** or similar (make sure to give you supervisor access). Keep a summary each week outlining what was agreed and what is to be done for the next meeting. Follow-on meetings can start by reviewing these minutes and action points.

If either the supervisor or the student is unable to keep a project-related appointment, the other party will be informed as soon as possible using #Slack. Naturally it will be rare for such cancellation to take place.

8. Project Finishing Date

The module will have an oral examination/demonstration in April/May. It is crucial to get started early and work on it each week, particularly in the implementation and dissemination stages.

Projects will be completed at this stage

Repeats will submit in August.

For capped independent projects where students want to complete certification over the summer, the project placement module is deemed to be complete in August. Initial Project marks may be updated to take account of certification for those not working in technology.

Project Marks will not officially be released until September irrespective of when you complete.

9. FINAL MARK - How does Work Placement, Project, and Certification come together?

As mentioned previously, the module title is called “ICT Skills Placement”. It incorporates Work Placement, Project, and Certification.

The Placement Co-ordinator (Joan Mangan) will request a job spec for your work placement or existing ICT role if upskilling, to advise if suitable or not.

Towards the end of placement, The Placement Co-ordinator will review your experience by first requesting your learning log (which might be recorded through a blog / vlog / competency document, etc.). This will be used to review your experience in preparation for a site visit and interview with your mentor. A result of pass / fail is recorded

If the spec is suitable and you completed sufficiently, this is recorded positively.

If the spec is unsuitable this is recorded negatively and you are recommended to complete the certification to maintain your indicative project grade.

Indicative project grades are given for the Project after correction in April/May.

You will have the option to complete certification. This is checked at the end of August

Final Marking for the module:

Placement / existing ICT	Certification completed successfully	Initial indicative project grade	Module Mark
Y	Y	>= PASS	Remains same
Y	N	>= PASS	Remains same
N	Y	>= PASS	Remains same
N	N	>= PASS	Reduced by a grade. (Cap of 70, floor of 40)
		FAIL (in April)	Review again in August
		FAIL (in August)	FAIL MODULE, don't graduate. MUST REGISTER TO REPEAT MODULE FOLLOWING YEAR (CURRENT COST PAYABLE BY STUDENT €1200 – reduced from €1675)

Appendices

APPENDIX 1: Project Proposal Outline

- Your project proposal must have a **title** (see below).
- Project Type (Word Based / Independent / Research)
- Project Category (Web App, Native Android, Hybrid, CI/CD PIPELINE, RESEARCH, Other)
- Your project proposal will be about **four or five hundred words long**.
- Please re-read your proposal when written, and amend it if necessary, to ensure that it is easy to read and makes sense to another person.
- You may wish to include a diagram or two.
- You should provide a list (with a title) of **technologies** that your project will employ, both software (including programming languages) and hardware.
- You should provide a list (with a title) of **tools** and frameworks that you intend to use.
- You should indicate the project **process** you intend to use: for example: whether you intend to implement your software all at once or as a series of production quality releases where more functionality is added as you go along.
- You should indicate other parties with whom you are collaborating or parties for whom the software is being written. These are sometimes collectively referred to as **stakeholders**.
- A title page is important to ensure that we can easily file and retrieve your report. Please provide a page at the front carrying the following information:
 - A title of your proposed project, for example: *Distributed Book Retrieval System* or whatever is appropriate.
 - The words *Project Proposal* below that in a smaller font.
 - Your name and student number
 - Your programme: Higher Diploma in Science in Computer Science.

APPENDIX 2: Video Guide

This is a simple video ideally 3-7 mins in duration, but we will allow up to a maximum of 10 minutes (The panel are instructed to stop playing at 10 minutes). It is a demonstration of the key features of your project. It acts as a backup in case you can't get the live demo working on presentation day. It also provides a convenient way to see the project working. It helps focus the mind on the creating a good presentation.

It is not expected to be a Hollywood production. Depending on the project you might:

- use a screen recording to do a walkthrough of the software implemented
- use a phone (in landscape mode) to record a video of some hardware you are controlling with software.
- Do a video recording highlighting the key elements of a research project

The video must have voice commentary (your voice) explaining what's happening. It must use your voice explaining what is happening. Typically, I would suggest start by introducing yourself "Hi my name is X" and my final project is called Y. One sentence description of the system. Then demo.

Most systems have registrations & logins, we take it for granted that this functionality is working at this stage. By all means, login/register if you wish (quickly) but don't labour the point and don't waste too much time in demonstrating how great your login/registration is. We are more interested in what makes your system different. Focus on the key aspects of your project.

If recording mobile, ideally check if your software has a trails feature, which makes it much easier to see the user interaction.

Video Software Recommendations:

Desktop:

You've already used **OBS** to record videos, so this is a fine way of recording the desktop.

Camtasia is the best screen recorder/editor on PC.

ScreenFlow is the best screen recorder/editor on Mac.

Demos are available (don't worry about watermarks on any software as long as they are not excessive)

Jing is free.

Loom is free.

There are others.

Mobile:

Built in screen recorders.

Camtasia & ScreenFlow record from your mobile device too.

Zoom

Android:

DU Recorder Screen Recorder

iOS:

RecordIt!::Screen Capture

Submitting

There are **two parts** to submitting:

URL

Upload your video to YouTube or similar (make it unlisted or public). Copy the URL. Paste it in the 'Online text' textbox. Highlight it and make it a hyperlink by clicking the link button and pasting the URL again.

FILE

Upload your video file to Moodle also in the File submission box - you can just drag and drop. It will most like be an mpeg4 encoded video (.mp4, .mkv)

Project Video

<https://tutors-design.netlify.com/talk/wit-hdip-comp-sci-2018-sept-onsite.netlify.com/topic-01-project/talk-5-project-handbook/project-handbook.pdf>

File submissions

Maximum file size: 2GB, maximum number of files: 3

File

ZoomBreak...

Online text

URL

<https://youtu.be/w8Uwy-9YXkY>

Save changes Cancel

When you click Save, your Moodle submission will look something like this.

Project Video

<https://tutors-design.netlify.com/talk/wit-hdip-comp-sci-2018-sept-onsite.netlify.com/topic-01-project/talk-5-project-handbook/project-handbook.pdf>

Submission status

Attempt number	This is attempt 1.
Submission status	Submitted for grading
Grading status	Not graded
Due date	Sunday, 31 May 2020, 11:55 PM
Time remaining	2 days 11 hours
Last modified	Friday, 29 May 2020, 12:11 PM
File submissions	 ZoomBreakoutRoomGuide.mp4 + 29 May 2020, 12:11 PM Export to portfolio

Submission comments
▶ [Comments \(0\)](#)

Online text



[Export to portfolio](#)

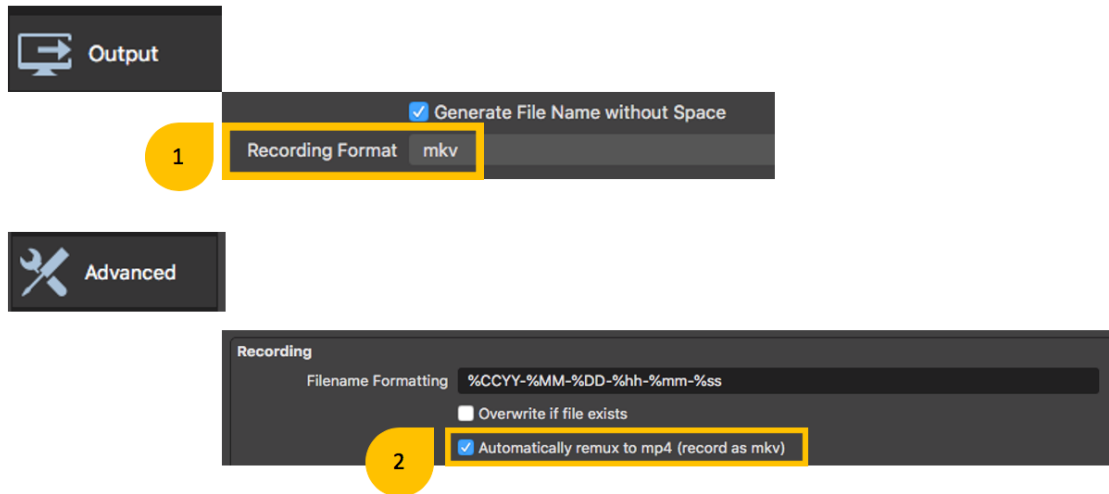
Edit submission

Remove submission

You can still make changes to your submission.

Additional:

Tip for recording in OBS



.mkv & .mp4 about 10 seconds after you press stop

APPENDIX 3: Presentation Guide

The week before your presentation, create your slides. These will be submitted to Moodle. It's a good idea to arrange & rehearse presentations a few days before with each other. Honest critiquing will help everyone improve.

You have up to **15 minutes to present**, followed by **5-10 minutes of Q&A**. So, this time will need to be managed well.

Planning your Presentations:

Part 1 – Introduction

1. Introduce yourself.
2. Give the name of the project and your short abstract (1 or sentences)

Part 2 - Sales Pitch (everything is great)

1. Demo what you've got
(This is ideally a live demo, but just in case it's good to have a recorded backup)

It's a good idea to use **real data** to help with understanding the system (i.e. don't use "test")

Part 3 - Academic Explanation (SUMMARY OF KEY POINTS OF YOUR REPORT)

1. i.e. how did you built it from start to finish? Analysis, design, plans, methodology, artefacts created along the way.
2. Summarise sections of your final report. What you learned. Key milestones. (demo videos). Problems encountered & solutions. Future development

Part 4. Wrap up

Any Questions + ask if anyone wants to try it out (live demo again)

Typical Timing

Part 1 & 2 will take 5 mins.

Part 3 & 4 will take 10 mins.

Q&A will take 5-10 mins

Tips

- Split up what you want to say and allocate a time budget to achieve it.
- Summarize and Visualize
- Use images instead of text where possible
- Short videos can also help
- A good yardstick is have no more than 7 points on the screen at any given time
- Max of 25 words on screen (there are exceptions)

JUNE 2020

“RUNNING SOCIAL”
Cloud Native App Development Pipeline
FINAL REPORT

NAME:

STUDENT ID:

LECTURER: COLM DUNPHY

MODULE: PROJECT

COURSE: Higher Diploma in Computer Science

Table of Contents – Samples

Table of Contents

Table of Contents

Declaration.....	2
Acknowledgments	3
Titles	4
Comm Go to page 22	4
Academic Title	4
Preface	5
Abstract.....	6
Table of Contents.....	7
Table of Figures	9
1 Introduction	10
1.1 Background	10
1.2 Problem Definition	10
1.3 Solution	11
1.4 Project Goal.....	11
2 Research and Analysis	12
2.1 Requirements analysis	12
2.1.1 Front End	12
2.1.2 Back End	13
2.1.3 CI/CD	14
2.2 Market analysis.....	15
2.3 Feature analysis	16
3 Modelling	17
3.1 Back End and Database	17
3.1.1 Data Modelling	17
4 Planning	20
4.1 Agile	20
4.2 Tools	20
4.3 Getting Started	20
5 Implementation	22
5.1 System Model.....	22
5.2 Feature Summary.....	22
5.3 Frontend and UI:.....	24
5.3.1 UI/System Walkthrough.....	25
5.4 Backend API.....	38

7

5.5 Deployment Infrastructure	40
5.6 Continuous Integration/Continuous Deployment	41
6 Reflection.....	46
6.1 Problems Encountered	46
6.2 Learning	48
6.3 Achievements	49
6.4 Future Development	50
8 Citations.....	51
9 References	53
Appendix A – Yarn Swap Logo Evolution	57
Appendix B – Wireframes	58
Appendix C – Jira Burnup Report by Sprint.....	63
Appendix D – Deployment Procedure	69
Appendix E – Ethics Questionnaire Responses	74

Table of Contents

1. Abstract	5
2. Introduction	6
3. Research and Analysis of tech options	7
3.1 Front-end options	7
3.1.1. Functionality	7
3.1.2. Design	8
3.2 Back-end/API options	9
3.2.1. API options	9
3.2.2. Authentication/Storage options	11
4. Modelling and Design	12
4.1. User types and user stories	12
4.2. Screen Mockups	15
4.3. Data modelling	20
4.4. System Diagram	24
4.5. Design	25
5. Development phases/project implementation	31
5.1. Building the React way	31
5.2. Development methodology/agile sprints	31
5.3 Key accomplishments and challenges	33
5.3.1. Sprint 1 - RAMP phase	33
5.3.2. Sprints 2 and 3 - Interim Report and development of static version of MVP	33
5.3.3. Sprint 4 - API, routing and consumer user stories	37
5.3.4. Sprint 5 - Creator user stories	42
5.3.5. Sprint 6 - Shipping new features, design, deployment and report	47
6. Conclusion	53
Bibliography	55
Appendices	58

Table of Contents

Declaration of Authenticity.....	i
Acknowledgements.....	i
Titles.....	i
Abstract.....	i
Important Links.....	i
Table of Figures.....	iv
Table of Tables.....	vi
1 Project Overview.....	1
1.1 Background to need for Spruuk.....	1
1.1.1 Client requirements.....	1
1.1.2 Vendor requirements.....	1
1.2 Solution overview – Spruuk.....	2
2 Competitive landscape.....	2
2.1 Houzz.....	2
2.2 MyFolio3.....	4
2.3 Contractor's Work.....	6
2.4 Summary.....	7
3 Analysis of project technologies.....	7
3.1 Front-end technologies.....	7
3.1.1 Flutter.....	7
3.1.2 Ionic.....	10
3.2 Back-end technologies.....	11
3.2.1 Firebase.....	11
3.2.2 Back4App.....	12
4 Design/Analysis.....	13
4.1 Methodology.....	13
4.2 Technology utilisation.....	14
4.2.1 Frontend/backend selection.....	14
4.2.2 Programming language.....	14
4.2.3 Integrated development environment.....	14
4.2.4 Source code management.....	15
4.2.5 File structure.....	15
4.2.6 State management.....	15
4.3 Database, models, and user experience design.....	17
5 Implementation.....	17
5.1 Sprint 1a: Research & Analysis.....	17
5.2 Sprint 1b: Design Planning.....	17
5.3 Sprint 2: Firebase Setup and User Authentication.....	17
5.4 Sprint 3: Create Read Update Delete (CRUD) of projects.....	18
5.5 Sprint 4: Search, favourite, and display projects on interactive maps.....	19
5.6 Sprint 5: CRUD for requests, responses, and messages.....	20
5.7 Sprint 6: Project reviews and profile updates.....	21
6 Reflection.....	21
6.1 What I have achieved.....	21
6.2 What I have learned.....	22
6.2.1 Flutter/Dart.....	22
6.2.2 Riverpod.....	22
6.2.3 App architecture.....	22
6.2.4 Plugin functionality.....	22
6.3 Problems encountered.....	22
6.3.1 Rapidly evolving platform.....	22
6.3.2 Difficulties using Riverpod.....	22
6.3.3 Using Google Sign-in but also setting up user type.....	23
6.3.4 Asynchronous functions and providers.....	23
6.3.5 Google Maps not working on web app.....	23
6.3.6 Device connection lost with use of Geolocator plugin.....	23
6.3.7 Continuity of profile image updates.....	23
6.4 Future development.....	24
6.4.1 Input variables and search terms related to project type.....	24
6.4.2 Administration user features, including consent approval.....	24
6.4.3 Searching by region.....	24
6.4.4 Saving search terms and one click filters.....	24
6.4.5 Prioritising project/request list displays based on proximity to user.....	24
6.4.6 Secondary vendor tagging.....	25
6.4.7 Reviews and analytics.....	25
6.4.8 Streamlining.....	25
7 Conclusion.....	25

8 References.....	26
Appendix.....	28
Database Design.....	28
Conceptual Database Design.....	28
Relations for Logical Data Model.....	28
Models.....	31
Project.....	31
User.....	32
Review.....	32
Request.....	33
Response.....	33
Message.....	33
User scenarios.....	34
Vendor.....	34
Client.....	39
Figma screenshot.....	43
Trello screenshot.....	44
Development phase diary.....	44
Sprint 2.....	44
Sprint 3.....	47
Sprint 4.....	50
Sprint 5.....	53
Sprint 6.....	55
Ethics Submission.....	55

Table of Contents

LIST OF FIGURES

LIST OF ABBREVIATIONS

1. INTRODUCTION

- 1.1 Background
- 1.2 Objectives
- 1.3 Motivation

2. PROJECT MANAGEMENT

- 2.1 Methodology
- 2.2 Project Board
- 2.3 Iteration 1
- 2.4 Iteration 2
- 2.5 Iteration 3
- 2.6 Iteration 4
- 2.7 Iteration 5
- 2.8 Changes to Plan
- 2.9 Project Access

3. DESIGN & ANALYSIS

- 3.1 Introduction
- 3.2 System Context Diagram
- 3.3 Container Diagram
- 3.4 Component Diagram

4. IMPLEMENTATION

- 4.1 Introduction
- 4.2 Component: cicd-common
- 4.3 Component: cicd-datasource
- 4.4 Component: cicd-streamsets
- 4.5 Template: cicd-postgres-to-greenplum
 - 4.5.1 Boilerplate Tests
 - 4.5.2 Supported Assertion Tests
- 4.6 Project: cicd-postgres-to-greenplum-reference
 - 4.6.1 Workflow Configuration
 - 4.6.2 CI/CD Stages
- 4.7 Project: cicd-shared-jobs
- 4.8 Docker: postgres-database
- 4.9 Docker: streamsets-datacollector

5. TESTING & VALIDATION

- 5.1 Testing Strategy
- 5.2 CI Automation
- 5.3 Code Quality

6. USER EXPERIENCE

- 6.1 Introduction
- 6.2 Create a New Project
- 6.3 Rename the Project
- 6.4 Get Access to the Code
- 6.5 Make Changes to The Code
- 6.6 Pipeline Monitoring

7. CONCLUSION

- 7.1 Reflection
- 7.2 Key Skills
- 7.3 Challenges
- 7.4 Future Work

8. REFERENCES

9. CITATIONS

10. APPENDIX A - DOCUMENTATION: CICD-COMMON

11. APPENDIX B - DOCUMENTATION: CICD-DATASOURCE

12. APPENDIX C - DOCUMENTATION: CICD-STREAMSETS

✓ Background

-  Current Features

✓ Objectives

-  Main Focus of the Project
-  Minor goals of the Project

✓ Methodology

-  Project Management
-  Programming Languages
-  Deployment Technologies
-  3rd Party Solutions/Technologies

✓ Project Plan

-  Phase 1 Inception
-  Phase 2 Development
-  Phase 3 Dissemination

✓ Development

-  Proof of Concept
-  Decouple UI from back end Server
- >  Building Container Images and Setup of Docker-compose
- >  Evaluating OpenFaas for Serverless functions
-  Refactor back end server to serverless functions
-  Final Testing and Deployment

✓ Project Evaluation

-  Main Project Objectives
-  Minor objectives of the project

Conclusion

Project URLs

-  Site Address
-  Source Code Repositories
-  Docker Images

Table of Contents

Declaration	
Acknowledgements	
Titles	
Commercial Title	
Academic Title	
Prefaces	
Abstract	
Keywords	
Figures	
Tables	
1.0 Introduction	
1.1 Background	
1.2 Scope	
1.3 Objectives	
1.4 User Requirements	
1.5 Github Repository	
1.6 Trello Board Kanban	
Go to page 18	
2.1 Use Case Diagram	
2.2 Entity Relation Diagram	
2.2.1 Entities	
2.2.1.1 User	
2.2.1.2 RoomBooking	
2.2.1.3 DeskBooking	
2.2.1.4 Desk	
2.2.1.5 Room	
2.2.1.5.1 Office	
2.2.1.5.2 Meeting	
2.2.1.5.3 Conference	
2.2.1.6 Chair	
2.2.1.7 Computer	
2.2.1.8 Monitor	
2.2.1.9 Docking Station	
2.2.1.10 Phone	
2.2.2 Relationships	
2.2.2.1 User-Deskbooking	
2.2.2.2 User-Roombooking	
2.2.2.3 Deskbooking – Desk	
2.2.2.4 Roombooking-Room	
2.2.2.5 Desk-Room	
2.2.2.6 Room - Mandatory Or	
2.2.2.7 Desk – Chair	
2.2.2.8 Desk – Computer	
2.2.2.9 Desk – Monitor	
2.2.2.10 Desk – Docking Station	
2.2.2.11 Desk – Phone	
2.3 UML Diagram	
2.3.1 User	
2.3.2 Desk Booking	
2.3.3 Room Booking	
2.3.4 Desk	
2.3.5 Room	
2.3.6 Room – Office	
2.3.7 Room – Meeting	
2.3.8 Room – Conference	
2.3.9 Chair	
2.3.10 Computer	
2.3.11 Monitor	
2.3.12 Docking Station	
2.3.13 Phone	

3.0 UX Mock-ups	
3.1 User Mock-ups	
3.2 Manager Mock-ups	
4.0 Alternative Applications	
4.1 Condeco	
4.2 Trimble	
4.3 Table of Comparison	
5.0 System Model	
5.1 Splash Screen	
5.2 Log In Screen	
5.3 Welcome – Launch Screen	
5.4 Initial Booking Options Screen	
5.5 List of Meeting/Conf Rooms and List of Offices	
5.6 Office List -> List of Available Desks in an Office	
5.7 Individual Desk Details	
5.8 List of Bookings	
5.9 Booking Confirmation	
5.10 Cancel A Desk Booking	
6.0 Technologies Utilised	
6.1 Native Android Application	
6.2 Kotlin Programming Language	
6.3 Google Firebase Authentication	
6.4 Google Firebase Realtime Database	
6.5 Google Firebase Cloud Functions	
6.5 Github & Sourcetree	
7. Application Iterations	
7.1 Iteration Plan	
7.2 Iterations	
7.3 Desk Request Version 1.0	
7.3.1 Iteration 1 – Splashscreen & Welcome	
7.3.2 Iteration 2 – Initial User Selection	
7.3.3 Iteration 3 – Add Office	
7.3.4 Iteration 4 – JSON Persistence	
7.3.5 Iteration 5 – Desk Details	
7.3.6 Iteration 6 – Booking Capability	
7.3.7 Iteration 7 – View Bookings	
7.3.8 Iteration 8 – Firebase User Authentication and Firebase Realtime Database	
7.3.9 Iteration 9 – Release 1 – Firebase Booking	
7.4 Desk Request Version 2.0	
7.4.1 Iteration 10 – User Settings	
7.4.2 Iteration 11 – View Bookings	
7.4.3 Iteration 12 – Release 2 - Booking Update and Delete	
7.5 Desk Request Version 3.0	
7.5.1 Iteration 13 – Room Booking	
7.5.2 Iteration 14 – Release 3 – Room Bookings Details	
7.6 Desk Request Version 4.0	
7.6.1 Iteration 15 – Release 4 - Email Confirmation	
7.7 Desk Request Version 5.0	
7.7.1 Iteration 16 – Release 5 – Bug Fixes	
8.0 Problems Encountered and Solutions Found	
8.1. Passing Dates between Activities	
8.2 RecyclerView View for a ArrayList in a Data Class	
8.3 Null Constructor Issue for Desk ArrayList Item	
8.4 Rooms Collection to the Firestore Realtime Database	
8.5 Kotlin Data Class Inheritance	
8.6 Email Booking Confirmation	
9. Reflection	
9.1 What I learned	
9.2 What I achieved	
9.3 Limitations and Future Possibilities	
9.4 Conclusion	
Appendices	
Bibliography	

Contents

1 Introduction

- 1.1 Devices
 - 1.1.1 Medical Scales
 - 1.1.2 Blood pressure monitor
- 1.2 Weight Management Service

2 Software

- 2.1 Initial consideration
- 2.2 Research
- 2.3 JAMStack
 - 2.3.1 Advantages
 - 2.3.2 Redwoodjs/Fauna/Vercel stack
 - 2.3.3 Svelte
- 2.4 JAMstack vs MEAN vs LAMP

3 Data Modelling

4 Software Review

- 4.1 Chosen platform - Svelte
- 4.2 Lambda Function
 - 4.2.1 Adding data to a patient
- 4.3 Sapper

5 Implementation

- 5.1 Components
- 5.2 Modal
- 5.3 Patient Dashboard
 - 5.3.1 Surgery Dashboard
 - 5.3.2 Cards with charts
 - 5.3.3 Navbar
 - 5.3.4 Navbar improvements
- 5.4 Frappe Charts
 - 5.4.1 Multiple lines
 - 5.4.2 Thresholds
 - 5.4.3 Zones
 - 5.4.4 Future charts
- 5.5 Firebase
 - 5.5.1 Firebase Realtime Database
 - 5.5.2 Firebase Realtime Database Rules
- 5.6 Email Alerts
 - 5.6.1 Firebase functions
 - 5.6.2 Implementation of Email alerts

- 5.6.3 Calculating the Change in weight
- 5.6.4 Healthmail
- 5.7 Json data to surgery
 - 5.7.1 Sending data to the Surgery patient management software
- 5.8 User levels
 - 5.8.1 Guest Level
 - 5.8.2 User Level - Surgery
 - 5.8.3 Administrator Level
- 5.9 Workflow using different methods
 - 5.9.1 Initial implementation
 - 5.9.2 Fixed Sidebar
 - 5.9.3 Future development of the Side panel
- 5.10 Admin Analytics
- 5.11 Firebase Storage
 - 5.11.1 Limitations and future development
- 5.12 Handling data in Svelte between components

6 Firebase Costings

- 6.1 Storage Costs
 - 6.1.1 Weight
 - 6.1.2 Bloodpressure readings
 - 6.1.3 Blood pressure Device
 - 6.1.4 Blood Pressure Device Admin
 - 6.1.5 Scales
 - 6.1.6 Scales Admin
 - 6.1.7 Patients
 - 6.1.8 Surgeries
 - 6.1.9 Users
- 6.2 Total
- 6.3 Downloads per day
- 6.4 Cloud functions
- 6.5 Cloud storage
- 6.6 Hosting

7 Future development work

- 7.1 Current issues

8 Conclusion/Reflection

1	Introduction	
1.1	Introduction To Document	4.4.4 Normal User Adds a Ringfort via the NMS Map Screen
1.2	Project Background	4.4.5 Overview of All Possible Screen flows.
1.3	Project Objectives	
1.4	Project Methodology	
1.5	Conclusion	
2	Project Requirements	
2.1	Introduction to Requirements	
2.2	Functional Requirements	
2.3	Non-Functional Requirements	
3	Analysis of Technology Options	
3.1	Introduction	
3.2	Frontend Technology Options	
3.2.1	Introduction	
3.2.2	Flutter	
3.2.3	React Native	
3.2.4	Ionic	
3.2.5	Head-To-Head Comparison	
3.2.6	Cross-Platform Mobile Frameworks Used Worldwide 2019-2021	
3.2.7	Conclusion	
3.3	Backend Technology	
3.3.1	Introduction	
3.3.2	Custom Node API (Application Programming Interface)	
3.3.3	Firebase Back End as a Service (BaaS)	
3.3.4	Conclusion	
3.4	Mapping Technology	
3.4.1	Introduction	
3.4.2	MapBox	
3.4.3	OpenStreetMaps (OSM)	
3.4.4	Google Maps	
3.4.5	Conclusion	
4	Design/Analysis	
4.1	Introduction	
4.2	Screen Mock-ups	
4.3	Data Modelling	
4.3.1	user Collection Model	
4.3.2	historicSites Collection Model	
4.3.3	historicSitesStaging Collection Model	
4.3.4	NMS-Ringforts Collection Model	
4.4	Process And Data Flow	
4.4.1	Introduction	
4.4.2	Admin User Adds/Updates a Ringfort directly via the Add/Update Screen.....	
4.4.3	Normal User Adds/Updates a Ringfort directly via the Add/Update Screen.....	
4.4.4	Normal User Adds a Ringfort via the NMS Map Screen	
4.4.5	Overview of All Possible Screen flows.	
4.5	Source Code Structure	
4.5.1	Introduction	
4.5.2	Firebase Access Classes	
4.5.3	General Helper Classes	
4.5.4	Data Model Classes	
4.5.5	State Management Provider Classes	
4.5.6	Screen Classes	
4.5.7	Widget Classes	
4.5.8	Main Class	
5	Development Phases	
5.1	Introduction	
5.1.1	Sprint 1	
5.1.2	Sprint 2	
5.1.3	Sprint 3	
5.1.4	Sprint 4	
5.1.5	Sprint 5	
5.1.6	Sprint 6/7/8/9	
6	Critical Self Analysis	
6.1	Introduction	
6.2	What I learned	
6.2.1	Flutter SDK And Dart Programming Language	
6.2.2	Agile Development, Project Planning and Re-planning	
6.2.3	Report Writing	
6.3	What I Achieved	
6.3.1	Building Cross Platform Mobile Application	
6.3.2	Making NMS Data Available to My Application	
6.4	Problems I encountered	
6.4.1	Proving the Application on an IOS Device	
6.4.2	State Management in the Application	
6.5	Possible Future Development	
7	Project Conclusion	
	Bibliography	
	Appendices	
	Appendix A – Hand Drawn Screens	
	Appendix B – Wireframe Design using Online Tool	
	Appendix C – User Stories	
	Appendix D – Original Sprint Plan Progress as at 07.02.2022	
	Appendix E – Updated Sprint Plan Progress as at 01.03.2022	
	Appendix F – Updated Sprint Plan as at 14.03.2022	
	Appendix G – Firebase Pricing	
	Appendix H – Trello Sprint Tracking	

Table of Contents

List of Tables

List of Figures

INTRODUCTION

Motivation.....

Background

Objectives.....

PROJECT MANAGEMENT.. Go to page 8

Project Methodology

Project Board

Code Repositories

Iteration 1.....

Iteration 2.....

Iteration 3.....

Iteration 4.....

Iteration 5.....

Iteration 6.....

Iteration 7.....

Iteration 8.....

DESIGN & ANALYSIS.....

Introduction

Component Diagram

Identification of Entities.....

Identification of Relationships.....

Identification and association of attributes with entities.....

Identification and association of attributes with relationships.....

Determination of Attribute Domains

Candidate Keys, Primary Keys, Alternate Keys

Logical Database Design

UML Diagram.....

Use Case Diagram

Level 1 Use Case

Level 2 Use Case

Process Flow Diagram

IMPLEMENTATION.....

Server – Python Flask API & Web-Scraper

Component Requirements

Research

Component Technologies.....

Server – Nest.js

Component Requirements

Research

Component Technologies.....

Client – Vue.js

Component Requirements

Research

Component Technologies.....

TESTING & VALIDATION.....

Testing Strategy

Server – Python Flask API & Web-Scraper

Server – Nest.js

Client – Vue.js

Code Quality.....

Server – Python Flask API & Web-Scraper

Server – Nest.js

Client – Vue.js

CONTINUOUS INTEGRATION (CI).....

USER EXPERIENCE

Introduction

Registration

User Verification

Home Page

Event Creation / Update Form.....

Event Poll Creation / Update Form

Event Poll Voting.....

Event Detail Page – Event Overview

Event Detail Page – Poll Overview

Event Detail Page - Event Itinerary.....

Event Detail Page – Scraped Information Table – Hidden.....

Event Detail Page – Scraped Information Table - Displayed ..

Event Detail Page – Chat Functionality.....

User Settings Page – Account Updates

DEVELOPER EXPERIENCE

Nest.js

Python Flask API & Web-Scraper

CONCLUSION

Reflection

Key Skills

Challenges

Future Work

REFERENCES

CITATIONS

APPENDIX A – Competitor Analysis – Feature List

APPENDIX B – Front-End Frameworks

Appendix C – 2021 Stackoverflow Developer Survey – Framework Usage Statistics

Appendix D – 2021 Stackoverflow Developer Survey – Developers with Experience in Back End Technology

Appendix E – 2021 Stackoverflow Developer Survey – Database Usage Statistics

APPENDIX 5: Declaration

I declare that the work which follows is my own, and that any quotations from any sources (e.g. books, journals, the internet) are clearly identified as such by the use of 'single quotation marks', for shorter excerpt and identified italics for longer quotations. All quotations and paraphrases are accompanied by (date, author) in the text and a fuller citation is the bibliography. I have not submitted the work represented in this report in any other course of study leading to an academic award.

Student..... Date

Work Place Mentor..... Date

APPENDIX 6: Academic Writing Workshops

To assist with your academic writing, MS-Word, **Mybib.com**, Zotero, and Mendeley are recommended for managing references and **citations**.

If you need help with academic writing you should attend the Academic Writing Workshops which are delivered remotely.

https://www.wit.ie/current_students/student_life_and_learning/semester-1-supports/#academic-writing

To watch a videos of a missed workshops:

[mailto:studentsupport.wd@setu.ie ?subject=missed workshop](mailto:studentsupport.wd@setu.ie?subject=missed%20workshop)

2023 Week 3 - Referencing, Paraphrasing & Citing (Harvard Style) is essential!

Sample Schedule

Academic Writing Workshops 2021-22				
Week	Day	Date	Time	Topic
Week 1	Monday	27th September	5.15pm	Academic writing Overview
	Friday	1st October	5.15pm	Academic writing Overview
Week 2	Monday	4th October	5.15pm	Essay Planning & Topic of Students Choice
	Friday	8th October	5.15pm	Essay Planning & Topic of Students Choice
Week 3	Monday	11th October	5.15pm	Referencing & Topic of Students Choice
	Friday	15th October	5.15pm	Referencing & Topic of Students Choice
Week 4	Monday	18th October	5.15pm	Reading Effectively
	Friday	22nd October	5.15pm	Reading Effectively
Week 5 (Revision)	Tuesday	26th October	5.15pm	Revision Workshop
	Wednesday	27th October	5.15pm	Revision Workshop
	Thursday	28th October	5.15pm	Revision Workshop
Week 6	Monday	1st November	5.15pm	Editing & Proof Reading, Assignment Checklists
	Friday	5th November	5.15pm	Editing & Proof Reading, Assignment Checklists
Week 7	Monday	8th November	5.15pm	Literary Reviews & Reading Effectively
	Friday	12th November	5.15pm	Literary Reviews & Reading Effectively
Week 8	Monday	15th November	5.15pm	Academic Toolkit & Time Saving Tips
	Friday	19th November	5.15pm	Academic Toolkit & Time Saving Tips
Week 9	Monday	22nd November	5.15pm	Referencing & Report Writing
	Friday	26th November	5.15pm	Referencing & Report Writing
Week 10 (Revision)	Tuesday	30th November	5.15pm	Revision Workshop
	Wednesday	1st December	5.15pm	Revision Workshop
	Thursday	2nd December	5.15pm	Revision Workshop (Possible Drop in Clinic)

APPENDIX 7: Sample Ethics Checklist for School of Science & Computing

1 * Are you a student on a **final year undergraduate** programme, a **taught postgraduate** programme, or are you conducting a **research project**?

☐ Final Year Undergraduate
☒ Taught Postgraduate
☐ Postgraduate Research Project
☐ Other Research Project

2 * What is the **working title** of your project?

MyApp - Full Stack Web Application

3 * Who are the project **supervisors/advisors/principal investigators**?

Colm Dunphy

4 * Does your project involve **human experimentation**? (e.g. clinical trials)

☐ Yes ☒ No

5 * Does your project involve **animal experimentation**?

☐ Yes ☒ No

Next Page >>

7 * Does your project involve the use of **human remains/cadavers/tissues/cells/biological fluids/embryos/foetuses**?

☐ Yes ☒ No

<< Previous Page Next Page >>

9 * Does your project involve the use of **animal tissues or biological fluids**?

☐ Yes ☒ No

<< Previous Page Next Page >>

11 * Does your project involve the **collection of rare or protected plants**?

☐ Yes ☒ No

12 * Does your project involve the generation or use of **genetically modified (GM) plant material**?

☐ Yes ☒ No

<< Previous Page Next Page >>

14 * Does your project involve the use of **ionising radiation**? (e.g. use of gamma ray spectrometry)

☐ Yes ☒ No

<< Previous Page Next Page >>

16 * Does your project involve the **collection of any new (or primary) data** from **individual people or groups**?

☐ Yes ☒ No

<< Previous Page

Next Page >>

21 * Does your project involve **persons under the age of 18 years** (i.e. minors), or **any vulnerable groups**? (e.g. prisoners, refugees, those in care, addiction service users, etc.)

☐ Yes ☒ No

22 * Does your project involve the use of **existing (or secondary) data**? (i.e. data originally collected for another purpose)

☐ Yes ☒ No

<< Previous Page

Next Page >>

25 * Before you submit the Ethics Checklist, you must **confirm all of the following**:

☒ I understand that the Ethics Checklist is a formal declaration.

☒ I have answered all questions on the Ethics Checklist carefully and truthfully.

☒

The supervisor/advisor (or principal investigator) for the project is present as the Ethics Checklist is being submitted, or they have given me explicit permission to submit it in their absence.

☒ I have had adequate ethics training and/or instruction prior to completing the Ethics Checklist.

☒ I understand, and agree to abide by, the general ethical principle of "do no harm" for this project.

☒ I will follow the instructions given in the Feedback Report.

26 * Authentication Code (ask your project supervisor/advisor for this code)

Enter Student Number:

Enter the Authentication Code below and click "Verify Code"

Note: If an INVALID authentication code is used then this submission is NULL and VOID

<< Previous Page

Submit Ethics Checklist